

Relative Exponential Periods on Polynomial Flags: Algebraic Independence over Fixed Boundary Fields

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Abstract

We construct rational differential systems for compact polynomial-chain exponential integrals, with polynomial twisted de Rham representatives and literal Stokes forcing from the fixed faces. For a real algebraic phase $q = \alpha x^a + \beta y^b + q_<$, where $a, b \geq 2$ are coprime and $q_<$ has lower weighted degree, we prove a single-chain comparison theorem under explicit Morse, critical-value independence, face-separation, and saddle-or-continuation hypotheses. The $(a-1)(b-1)$ monomial moments are algebraically independent over the fixed-face function field, and their values are algebraically independent over the corresponding face-value field at every nonzero algebraic parameter. Every polynomial amplitude has a unique Laurent Stokes remainder; algebraicity of its value over the face-value field is equivalent to polynomial twisted exactness.

The proof combines global Picard–Lefschetz geometry, a Fourier order-versus-degree irreducibility argument, direct simple-spectrum formal diagonalization, and an elementary stable-lattice obstruction to boundary adjoint factors. An actual one-column criterion converts these facts into algebraic independence. We verify both the original elliptic triangle and a rank-six triangle with phase $x^4 + y^3 + xy - y$. The boundary fields contain only the specified faces. No genericity assertion, several-chain comparison, or general formal-completeness theorem is claimed.

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1 Introduction and main comparison theorem

We study actual compact integrals and their polynomial Stokes certificates, not an unspecified completion of a formal period algebra. The basic polynomial boundary construction is general; the comparison theorem below verifies its actual-column criterion for a weighted-plane class under explicit hypotheses. The elliptic and rank-six examples give two complete applications.

The algebraic changes of variables, linearity, and twisted Stokes identities used here belong to the exponential Kontsevich–Zagier framework [6, 5]. The conclusions below concern one actual column over its fixed boundary field. They do not assert a presentation of all formal exponential-period relations. The functional and numerical independence proofs do not use the companion formal-comparison theorems or Paper IV.

Structure and previous version. Section 2 constructs actual systems and their E -function coordinates. Section 3 proves the general one-column criterion. The later sections verify its hypotheses using the geometry, Fourier irreducibility, and the fixed-face obstruction, and then specialize at nonzero algebraic parameters. Section 11 retains the explicit continuation needed for the original triangle. The pre-integration version [8], including its separate observable-Lefschetz, Hodge, face-lattice, and multicolumn discussions, is archived unchanged. Those additional discussions are not inputs to the proof here. The earlier unrestricted endpoint-complete claim remains withdrawn; it is not restored by this theorem.

Revision. The 6 October 2026 revision integrates the source-checked single-chain theorem and its stable-lattice proof. The manuscript’s author list and original month-year date are retained.

Throughout $k = \overline{\mathbb{Q}}$, $k_{\mathbb{R}} = k \cap \mathbb{R}$, $F_0 = k(z)$, $F = \mathbb{C}(z)$. For a polynomial phase q and a compact real polynomial two-chain D ,

$$I_D(\omega)(z) = \int_D e^{zq} \omega, \quad d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

$K_{\partial,0}$ is the differential field over F_0 generated by the polynomial-amplitude integrals over the fixed faces of D and the exponentials at their endpoints; $K_{\partial} = \mathbb{C}K_{\partial,0}$; $K_{\partial,\xi}$ is the field generated over k by the values at ξ . For a face e with parametrization ϕ_e put $p_e = q \circ \phi_e$ and $d_e = \deg p_e$.

Definition 1.1 (The class (W)). Fix coprime integers $a, b \geq 2$ and give x, y the weights $\text{wt}(x) = b$, $\text{wt}(y) = a$. A phase is of class (W) if

$$q = \alpha x^a + \beta y^b + q_<, \quad \alpha\beta \neq 0, \quad q_< \in k_{\mathbb{R}}[x, y] \text{ of weighted degree } < ab,$$

with $\alpha, \beta \in k_{\mathbb{R}}$. Put $\mu = (a-1)(b-1)$ and $B = \{x^i y^j : 0 \leq i \leq a-2, 0 \leq j \leq b-2\}$.

The elliptic phase $y^2 - x^3 + 3x$ is of class (W) with $(a, b) = (3, 2)$, $\mu = 2$.

Hypotheses. Let q be of class (W) and D a polynomial two-chain over $k_{\mathbb{R}}$: a finite k -linear combination of pushforwards of $[0, 1]^2$ by maps in $k_{\mathbb{R}}[u, v]^2$. This arithmetic restriction is used in (iii).

- (M) *Morse.* q has μ distinct critical values $c_1, \dots, c_\mu$ (equivalently: all critical points are nondegenerate with distinct values).
- (Q) *Independent critical values.* $c_2 - c_1, \dots, c_\mu - c_1$ are linearly independent over $\mathbb{Q}$.
- (F) *Faces.* For every face e with $d_e \geq 2$, the set $\{c_i - c_j : i \neq j\}$ is not contained in $\{u - v : u, v \text{ critical values of } p_e\}$.
- (S) *Saddle.* Some real critical point P_0 of q with $\det \text{Hess } q(P_0) < 0$ lies off the image of ∂D , the multiplicity w of D at P_0 is nonzero, and $c := q(P_0)$ is neither the value of q at a vertex of D nor a critical value of any p_e on its parameter interval.

For $\mu = 2$, (Q) is automatic.

Theorem 1.2 (Single-chain comparison). *Let q be of class (W) satisfying (M) and (Q), and let D be a polynomial two-chain over $k_{\mathbb{R}}$ satisfying (F) and either (S) or (S^*) below. Put $M_m = I_D(m dx \wedge dy)$ for $m \in B$.*

- (i) *The classes of $m dx \wedge dy$, $m \in B$, form an F_0 -basis of H_q^2 , and a $k[z, z^{-1}]$ -basis of the lattice of polynomial forms. Every $r \in k[x, y]$ has a unique remainder and a polynomial Stokes certificate*

$$r = \sum_{m \in B} a_{r,m}(z) m + L_x P_r + L_y Q_r, \quad a_{r,m} \in k[z^{-1}], \quad P_r, Q_r \in z^{-1} k[z^{-1}][x, y],$$

with $L_x = \partial_x + zq_x$, $L_y = \partial_y + zq_y$.

- (ii) *The μ functions M_m are algebraically independent over K_{∂} .*
- (iii) *For every $\xi \in k^\times$ the μ numbers $M_m(\xi)$ are algebraically independent over $K_{\partial, \xi}$.*
- (iv) *For $r \in k[x, y]$ and $\xi \in k^\times$, the value $I_D(r dx \wedge dy)(\xi)$ is algebraic over $K_{\partial, \xi}$ iff it lies in $K_{\partial, \xi}$ iff $r dx \wedge dy \in (d + \xi dq) \Omega_{k[x, y]/k}^1$.*

Remark 1.3 (The continuation alternative). Hypothesis (S) is used only in Lemma 8.1. It may be replaced by:

- (S^*) on some interval $e_0 \subset \mathbb{R}$ free of critical values of q and of the p_e , of vertex values and of 0, the density germ ρ of $I_D(dx \wedge dy)$ continues analytically along a path to a punctured neighbourhood of some c_i , where its variation around c_i is a nonzero constant multiple of the vanishing period $\int_{\nu_i(t)} dx \wedge dy / dq$.

Lemma 11.1 verifies (S^*) for the original elliptic triangle: its saddle $(1, 0)$ lies outside the triangle, so (S) itself fails. Section 11 verifies (F) and derives its functional and numerical comparison from Theorem 1.2.

Corollary 1.4 (Polygons). *Let $\min(a, b) \geq 3$, let q be of class (W) with (M), (Q), and let D be a real polygonal chain (all faces straight segments) with vertices in $k_{\mathbb{R}}^2$. Then (F) holds automatically. If moreover the coefficients of q and the vertices lie in a field $k_0 \subset k_{\mathbb{R}}$ over which $[k_0(c_i) : k_0] \geq \max(a, b)$ for all i , then (S) holds as soon as some real saddle of q lies in the interior of D with nonzero multiplicity.*

Proof. A straight face has $d_e \leq \max(a, b)$, so p_e has at most $\max(a, b) - 1 < \mu$ critical values and at most $(\mu - 1)(\mu - 2) < \mu(\mu - 1)$ nonzero differences, whereas under (Q) the $\mu(\mu - 1)$ numbers $c_i - c_j$ are pairwise distinct. Vertex values lie in k_0 and critical values of $p_e \in k_0[s]$ have degree at most $d_e - 1 < \max(a, b)$ over k_0 , so neither can equal c . $\square$

Example 1.5 (A rank-six instance). For

$$q = x^4 + y^3 + xy - y, \quad D = \text{conv}\{(0, -1), (1, -1), (\tfrac{1}{2}, 0)\},$$

the six functions $\int_D x^i y^j e^{zq} dx dy$ ($0 \leq i \leq 2$, $0 \leq j \leq 1$) are algebraically independent over the field of the three edges, and so are their values at every nonzero algebraic ξ over the edge-value field. See §10.

Differential-module convention. For a system $Y' = EY$ we use its coefficient module: a column of basis elements e satisfies $\partial e = Ee$. Solutions are differential homomorphisms out of this module, so the solution functor is contravariant. Its horizontal representation is dual to the solution representation. The natural trace duality identifies the corresponding endomorphism and trace-free endomorphism representations. All module embeddings below are under the covariant horizontal-fibre equivalence with these identifications.

2 Polynomial representatives and literal boundary forcing

Put $k = \overline{\mathbb{Q}}$ and $F_0 = k(z)$. Let D be a finite algebraic-linear combination of pushforwards of $[0, 1]^2$ by polynomial maps with real algebraic coefficients. Let $q \in k[x, y]$. For a polynomial two-form ω , write

$$I_D(\omega) = \int_D e^{zq} \omega.$$

When coefficients of ω lie in F_0 , interpret this as a meromorphic function of z by linearity. For each oriented face parametrization $\phi_e : [a_e, b_e] \rightarrow \mathbb{A}^2$, put $p_e = q \circ \phi_e$. The actual boundary field $K_{\partial, 0}$ is the differential field over F_0 generated by all functions

$$\int_{a_e}^{b_e} h(t) e^{z p_e(t)} dt, \quad h \in k[t],$$

and the exponential functions at their algebraic endpoints. Only the fixed faces are used; this does not mean the field of all compact line periods.

Let d differentiate only in x, y , and set

$$d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x, y]/F_0}^2 / d_z \Omega_{F_0[x, y]/F_0}^1.$$

Theorem 2.1 (Polynomial-representative boundary system). *The space H_q^2 is finite dimensional. There are forms $\omega_i = r_i(x, y) dx \wedge dy$, with $r_i \in k[x, y]$, whose classes form an F_0 -basis, a matrix $A \in M_n(F_0)$, and polynomial-in- x, y one-forms $\beta_i \in \Omega_{F_0[x, y]/F_0}^1$ such that*

$$q\omega_i = \sum_j A_{ij}(z)\omega_j + d_z\beta_i. \quad (2.1)$$

The actual integral column $U_i = I_D(\omega_i)$ satisfies

$$U' = AU + b, \quad b_i = \int_{\partial D} e^{zq}\beta_i \in K_{\partial, 0}. \quad (2.2)$$

Every polynomial two-form ω has an identity

$$\omega = \sum_i a_i(z)\omega_i + d_z\beta, \quad (2.3)$$

where $a_i \in F_0$ and β is polynomial in x, y . Consequently

$$I_D(\omega) = \sum_i a_i(z)U_i + \int_{\partial D} e^{zq}\beta. \quad (2.4)$$

For any finite set of amplitudes, these functions belong to a finite closed rational first-order system whose coordinates are actual E -functions. No spatial denominator is introduced. Parameter denominators may occur.

Proof. The polynomial Gauss–Manin complex for $q : \mathbb{A}^2 \rightarrow \mathbb{A}_t^1$ is

$$(\Omega_{k[x, y]/k}^{\bullet+2}[\partial_t], d - dq \wedge \partial_t),$$

and its cohomology is holonomic [9, Proposition 9.1]. The Fourier substitution $\partial_t \mapsto -z$, $t \mapsto \partial_z$ turns its differential into $d + z dq \wedge$. Fourier transform is exact on Weyl modules, and localization to $k(z)$ is exact. The degree-zero cohomology of the localized transformed complex is therefore precisely H_q^2 , which is finite dimensional over $k(z)$ by holonomicity. This uses neither cohomological tameness nor the rank and lattice conclusions stated later in [9]. In particular, no identification with a Milnor number is asserted here.

The monomial two-forms with coefficients in k span the quotient over F_0 , so a finite subset is a basis. The connection on the quotient is

$$\nabla_z[\omega] = [\partial_z\omega + q\omega],$$

because

$$[\partial_z + q, d + z dq \wedge] = dq \wedge - dq \wedge = 0.$$

Expressing $q\omega_i$ in the basis proves (2.1); expressing an arbitrary class proves (2.3).

For a one-form polynomial in the spatial variables, there is the literal identity

$$e^{zq}d_z\beta = d_{x, y}(e^{zq}\beta).$$

Stokes on the original parametrized compact chain proves (2.2) and (2.4). Each pullback of β to a face is a finite F_0 -linear combination of polynomial one-forms in its parameter. Thus every forcing term is an actual face integral with coefficients in F_0 .

To see finiteness on the boundary, if $\deg p = d \geq 2$, division by $\partial_t + zp'(t)$ reduces every polynomial amplitude to degree at most $d - 2$. The leading degree of $(\partial_t + zp')h$, for $h \neq 0$, is $\deg h + d - 1$, with nonzero leading coefficient in F_0 . The remainder is therefore unique. Integration of the exact part gives only endpoint exponentials. Multiplication by p and the same division give a rational first-order system for the reduced moments and the endpoint exponentials. For $d = 1$ the twisted operator is surjective on polynomials; for $d = 0$ ordinary polynomial antiderivatives suffice. In both cases only endpoint terms remain. For finitely many faces, this supplies a finite boundary vector; its fraction field is $K_{\partial,0}$.

The integral coordinates are entire and have algebraic Taylor coefficients. After pullback to the square they have the form $\int_{[0,1]^2} P(u,v) e^{zQ(u,v)} du dv$. For the first m Taylor coefficients, one denominator of the form

$$d^{m+1} \text{lcm}(1, \dots, Cm + C_0)^2$$

suffices, with fixed integers d, C, C_0 ; all conjugates grow exponentially. The same argument in one variable handles the faces. Together with the constructed differential systems, these are the E -function conditions. $\square$

Proposition 2.2 (Actual affine comparison under a change of representatives). *Suppose $\tilde{\omega} = T(z)\omega + d_z\gamma$, with $T \in \text{GL}_n(F_0)$. Put $g = \int_{\partial D} e^{zq}\gamma$. Then the actual columns satisfy*

$$\tilde{U} = TU + g,$$

and their systems are related by

$$\tilde{A} = (T' + TA)T^{-1}, \quad \tilde{b} = Tb + g' - \tilde{A}g.$$

In particular, this is a comparison of the actual affine solution objects, not a statement only about their homogeneous composition factors.

Proof. The first equality is Stokes, and differentiating it proves the other two. If two primitives give the same form identity, their difference is d_z -closed and its integral over ∂D is zero by the same Stokes identity. Thus primitive choices do not introduce an arbitrary constant. $\square$

After adjoining complex constants, write $F = \mathbb{C}(z)$, $K_{\partial} = \mathbb{C}K_{\partial,0}$, and work in a field $\mathcal{M}$ of meromorphic germs containing the actual functions. Integration modulo the actual boundary field gives a canonical differential map

$$K_{\partial} \otimes_{F_0} H_q^2 \longrightarrow \mathcal{M}/K_{\partial}, \quad [\omega] \longmapsto I_D(\omega) \bmod K_{\partial}. \quad (2.5)$$

Here $\mathcal{M}/K_{\partial}$ is used as a differential vector space, not as a quotient field or ring. This map need not be injective for arbitrary chains. The next criterion proves injectivity rather than building it into a completion definition.

3 An actual-solution comparison criterion

Let $\mathcal{B}$ be a finite rational boundary differential system, let L_{∂}/F be a Picard–Vessiot field for that system containing its actual solution coordinates, and let $\mathcal{N} \subset L_{\partial}$ be their finite-dimensional F -linear differential span. Include 1 when needed and put $K_{\partial} = F(\mathcal{N}) \subset L_{\partial}$. All fields are embedded in a common no-new-constants differential overfield. Let $\mathcal{T}_{\partial}$ be the tensor category generated by $\mathcal{B}$, allowing subquotients and duals. Its objects are precisely the differential modules trivialized by L_{∂} [10, Theorem 1.5.2].

Lemma 3.1 (Differentially finite elements of a Picard–Vessiot field). *Every finite-dimensional F -differential subspace of L_∂ belongs to $\mathcal{T}_\partial$.*

Proof. Let R be the Picard–Vessiot ring. If $f \in L_\partial$ satisfies a scalar linear equation over F , all its Galois translates satisfy that equation. Their constant span is finite dimensional. After trivializing the torsor, a rational function with finite-dimensional translation orbit has no pole: the union of the polar loci of a finite basis is a proper closed subset invariant under all translations, whereas the group acts transitively on itself. This remains valid component by component if the group is disconnected. Thus $f \in R$; see the locally finite argument in [10, proof of Proposition 1.3.33].

Write $R = F[Y_{ij}, (\det Y)^{-1}]$ for a boundary fundamental matrix. A bounded span of monomials in Y_{ij} and $(\det Y)^{-1}$ is stable under differentiation, since $Y' = A_\partial Y$ and $((\det Y)^{-1})' = -\operatorname{tr}(A_\partial)(\det Y)^{-1}$. Such a span is a quotient of finite sums of tensor constructions in the boundary module and its determinant dual. Every finite-dimensional F -differential subspace of R lies in one of these spans. Closure under subquotients and the Picard–Vessiot tensor equivalence [10, Theorem 1.5.2] prove the claim. $\square$

Lemma 3.2 (Regular affine cocycles). *Let G be an algebraic group over an algebraically closed field C_0 of characteristic zero, with reductive identity component, and let V be a finite-dimensional rational representation. Every regular cocycle $c(gh) = c(g) + gc(h)$ has the form $c(g) = v - gv$ for some $v \in V$.*

Proof. Let g act on $V \oplus C_0$ by $g(w, t) = (gw + tc(g), t)$. Complete reducibility supplies an invariant lift $(v, 1)$ of 1 in the trivial quotient, giving the assertion. Complete reducibility here includes disconnected groups: average a G° -equivariant splitting over the finite component group [7, Corollary 22.43]. Equivalently, this is the vanishing of algebraic $H^1(G, V)$ in [7, Proposition 15.15]. $\square$

Theorem 3.3 (Actual one-column comparison). *Suppose an actual column U satisfies $U' = AU + b$, with $b_i \in \mathcal{N}$, and $n \geq 2$. Assume:*

1. *the homogeneous Galois group over F contains SL_n ;*
2. *its adjoint module $\operatorname{End}^0(A)$ does not belong to $\mathcal{T}_\partial$;*
3. *at least one coordinate of U is not in $\mathcal{N}$.*

Then $U_1, \dots, U_n$ are algebraically independent over $K_\partial \subset L_\partial$. In the setting of Theorem 2.1, (2.5) is injective, and its image is the span of the actual polynomial-amplitude periods modulo K_∂ .

Proof. Let $E = \mathcal{N} + \sum_i F U_i$. The differential module $E/\mathcal{N}$ is a quotient of the coefficient module associated with A . Use the coefficient module with basis e_i and $\partial e_i = \sum_j A_{ij} e_j$; the map $e_i \mapsto U_i \bmod \mathcal{N}$ is a differential map. Its horizontal representation is dual to the solution representation of $Y' = AY$, so it is irreducible and has the same adjoint tensor module. By the third assumption, the quotient is nonzero and hence is that entire irreducible module.

If all U_i belonged to L_∂ , Lemma 3.1 would put E , then $E/\mathcal{N}$, and then its adjoint module in $\mathcal{T}_\partial$, contradicting the second assumption. Thus $U \notin L_\partial^n$.

In a joint Picard–Vessiot extension for A and the boundary system, the kernel of restriction to L_∂ maps to a normal subgroup H of the homogeneous group $G \subset \operatorname{GL}_n$. A normal subgroup of a group containing SL_n either contains SL_n or has trivial projective image. Indeed, if $H \cap \operatorname{SL}_n$ is not all of SL_n , it is finite central. For each $h \in H$, the commutator map from the connected group SL_n to this finite intersection is constant and equals the identity at 1. Thus h centralizes SL_n and

is scalar. This also excludes a hidden noncentral finite projective image. In the latter, projectively trivial case, H acts trivially on $\text{End}^0(A)$, so the adjoint module is trivialized by L_∂ , a contradiction. Therefore the homogeneous group after base change to L_∂ , and hence to K_∂ , contains SL_n .

Let Y be a fundamental matrix over a joint field for the inhomogeneous system over K_∂ . Its affine Galois group is a subgroup of $\mathbb{C}^n \rtimes G_{K_\partial}$, acting by $\sigma(U) = U + Yc_\sigma$. Its translation kernel is an invariant vector subspace, hence either 0 or $\mathbb{C}^n$. If it is $\mathbb{C}^n$, translations make the orbit of U Zariski dense in $\mathbb{A}^n$. If it is zero, the remaining regular algebraic cocycle of the homogeneous group is a coboundary by Lemma 3.2. Here reductivity of its identity component follows also directly: a unipotent radical would lie in SL_n and be normal under SL_n , which has no nontrivial connected normal unipotent subgroup. Explicitly, writing $\sigma(Y) = Yg_\sigma$, the cocycle can be written $c_\sigma = g_\sigma c - c$; hence $U - Yc$ is fixed. Thus

$$U = a + Yc, \quad a \in K_\partial^n, \quad c \in \mathbb{C}^n.$$

The vector c is nonzero, since otherwise $U \in K_\partial^n$. The SL_n -orbit of a nonzero vector is $\mathbb{A}^n \setminus \{0\}$, again dense. In either case a polynomial over K_∂ vanishing at U vanishes on a dense affine orbit and is zero. Linear independence modulo the boundary field, and then the last assertion by normal-form reduction, follow immediately. $\square$

Remark 3.4. The third condition concerns only rational *linear* combinations of a finite actual boundary vector. It is strictly weaker than the desired nonmembership in the boundary fraction field. The proof is what upgrades this linear test to the fraction-field statement. A nonzero abstract cohomology class alone would not suffice.

4 Normal form for the class (W)

Lemma 4.1. *Let q be of class (W) (coprimality is not used here).*

- (a) *Every $r \in k[x, y]$ can be written $r = \sum_{m \in B} a_m(z)m + L_x P + L_y Q$ with $a_m \in k[z^{-1}]$ and $P, Q \in z^{-1}k[z^{-1}][x, y]$.*
- (b) *If $\sum_{m \in B} a_m m = L_x P + L_y Q$ with $a_m \in F_0$ and $P, Q \in F_0[x, y]$, then all $a_m = 0$.*
- (c) *Consequently B is an F_0 -basis of H_q^2 and a $k[z, z^{-1}]$ -basis of $G = \Omega^2[z, z^{-1}]/d_z \Omega^1[z, z^{-1}]$. In this basis $\nabla_z[m] = [qm] = \sum_{m'} A_{mm'}[m']$ with*

$$A = A_0 + A_1 z^{-1} + \cdots + A_s z^{-s} \in M_\mu(k[z^{-1}]),$$

and A_0 is the matrix of multiplication by q on $k[x, y]/(q_x, q_y)$ in the basis B . Under (M) its eigenvalues are $c_1, \dots, c_\mu$, each simple.

The same statements hold with k replaced by $\mathbb{C}$, and the reduction and independence arguments also apply after specialization at every $z = \xi \neq 0$.

Proof. (a) Let $x^m y^n$ be a monomial of r , with coefficient γ , of maximal weight among those not in B . If $m \geq a - 1$ put $v = \gamma x^{m-a+1} y^n / (a\alpha z)$. Then $L_x v = \gamma x^m y^n + (\gamma/a\alpha) q_{<,x} x^{m-a+1} y^n + v_x$, and the last two terms have weight strictly below $\text{wt}(x^m y^n)$ because $\text{wt}(q_{<,x}) < ab - b$. Replace r by $r - L_x v$. If $m < a - 1$ then $n \geq b - 1$ and one uses L_y symmetrically. The maximal weight of a monomial outside B , with its multiplicity among the finitely many monomials of that weight, strictly decreases, so the process stops.

(b) Among all pairs (P, Q) representing the given combination choose one minimizing $\varpi = \max(\text{wt } P + ab - b, \text{wt } Q + ab - a)$. Let P^{top} and Q^{top} be the components of P and Q of weights $\varpi - ab + b$ and $\varpi - ab + a$ (one of them may be zero). The weight- ϖ component of $L_x P + L_y Q$ is $z(a\alpha x^{a-1} P^{\text{top}} + b\beta y^{b-1} Q^{\text{top}})$. If it is nonzero it is a nonzero element of the monomial ideal (x^{a-1}, y^{b-1}) , which cannot equal the weight- ϖ component of a combination of monomials of B . Hence it vanishes, and since x^{a-1}, y^{b-1} is a regular sequence, $P^{\text{top}} = b\beta y^{b-1} R$, $Q^{\text{top}} = -a\alpha x^{a-1} R$. As L_x and L_y commute, $(P - L_y(R/z), Q + L_x(R/z))$ represents the same element and has smaller ϖ unless $P = Q = 0$. So $P = Q = 0$ and all $a_m = 0$.

(c) The first two assertions follow from (a) and (b). In (a), the terms of order z^0 give $r \equiv \sum a_m(\infty)m$ modulo (q_x, q_y) , and the argument of (b) with z removed shows B is a basis of the Jacobian algebra; this identifies A_0 . That algebra is the product of its local algebras at the critical points, on each of which q has the single eigenvalue $q(P)$, with multiplicity the local Milnor number. $\square$

5 Period calculus and irreducibility of H_q^2

Let $S = \mathbb{C} \setminus \{c_1, \dots, c_\mu\}$, $C_t = q^{-1}(t)$. The arguments below use three geometric facts.

- (G1) q is a locally trivial fibration over S ; every fibre of q is reduced and irreducible; for $t \in S$, $H_1(C_t; \mathbb{C})$ has dimension μ and its intersection form is nondegenerate.
- (G2) The local monodromy T_i around c_i is the Picard–Lefschetz transvection $\delta \mapsto \delta \pm \langle \delta, \nu_i \rangle \nu_i$, and the vanishing period $\int_{\nu_i(t)} dx \wedge dy/dq$ is holomorphic at c_i with a nonzero value there; in particular $\nu_i \neq 0$ and $T_i \neq 1$.
- (G3) $\pi_1(S)$ acts irreducibly on $H_1(C_t; \mathbb{C})$.

Proposition 5.1 (Geometry of the class (W)). *For q of class (W) satisfying (M), (G1)–(G3) hold.*

Proof. Fibres. The top weighted form of $q - t$ is $\alpha x^a + \beta y^b$, which is irreducible in $\mathbb{C}[x, y]$ because $\gcd(a, b) = 1$. The top form of a product is the product of the top forms, so $q - t$ is irreducible; in particular every fibre is reduced and irreducible.

Tameness. Put $R = \max(|x|^{1/b}, |y|^{1/a})$, $X = x/R^b$, $Y = y/R^a$. Then $R^{-(ab-b)} q_x = a\alpha X^{a-1} + o(1)$ and $R^{-(ab-a)} q_y = b\beta Y^{b-1} + o(1)$ uniformly on $\max(|X|, |Y|) = 1$ as $R \rightarrow \infty$, and one of the two leading terms has modulus bounded below there. Hence $\|\nabla q\| \rightarrow \infty$ at infinity, so q is tame in the sense of Broughton [4, Definition 3.1]: $\|\partial q\|$ is bounded away from 0 outside a compact neighbourhood of the critical points. For tame polynomials, [4, p. 230] shows that q is locally trivial over any region of the target containing no critical value, and [4, Theorem 1.2 and p. 231] gives $H_1(C_t; \mathbb{Z}) \cong \mathbb{Z}^{\mu'}$ for a regular value t , where μ' is the sum of the Milnor numbers at the critical points; by Lemma 4.1(c), $\mu' = \dim \mathbb{C}[x, y]/(q_x, q_y) = \mu$.

Local model and stability. Write

$$f_\epsilon(X, Y) = \alpha X^a + \beta Y^b + \sum_{bi+aj < ab} q_{ij} \epsilon^{ab-bi-aj} X^i Y^j.$$

Every exponent of ϵ is a positive integer, so this is a holomorphic deformation of $f_0 = \alpha X^a + \beta Y^b$. For $\epsilon \neq 0$ it equals $\epsilon^{ab} q(\epsilon^{-b} X, \epsilon^{-a} Y)$; its critical points and values are $(\epsilon^b x_i, \epsilon^a y_i)$ and $\epsilon^{ab} c_i$. Choose a Milnor ball $\overline{B}_r$ for f_0 , and choose $\eta > 0$ so that its levels $|t| \leq 2\eta$ are transverse to ∂B_r .

Compactness and C^1 convergence preserve this transversality for f_ϵ with $|t| \leq \eta$ when $|\epsilon|$ is sufficiently small. Shrink the parameter disk so that all critical points lie in $B_{r/2}$ and all critical values have modulus less than $\eta/2$.

At the fixed value $t_* = \eta$, the family

$$\{(\epsilon, X, Y) : f_\epsilon(X, Y) = t_*, (X, Y) \in \overline{B_r}\} \longrightarrow \{\epsilon : |\epsilon| < \epsilon_0\}$$

is a proper submersion, including on its boundary. The boundary form of Ehresmann's theorem therefore identifies its fibres with the Milnor fibre of f_0 . This is also the perturbation comparison in [1, §2.1, pp. 29–31]. For a fixed small nonzero ϵ , the local proper submersion over the disk minus the critical values transports this identification to every regular level there. Put $S_t = f_\epsilon^{-1}(t) \cap \overline{B_r}$ and $C_t = f_\epsilon^{-1}(t)$. By [1, Theorem 2.1, p. 31] a distinguished set of vanishing cycles $\Delta_1, \dots, \Delta_\mu$ is a basis of $H_1(S_t; \mathbb{Z})$. For a plane-curve singularity the kernel of the intersection form has dimension $r_0 - 1$, where r_0 is the number of branches [1, §3.6, p. 100]. The binomial germ f_0 is irreducible, so this form is nondegenerate; compare also [1, p. 411, Example 2].

Compatible local and global transport. Tameness is preserved by the fixed nonzero rescaling. Over a relatively compact target neighborhood U whose closure avoids the critical values, $\|\nabla f_\epsilon\|$ is bounded below: use tameness outside a compact set and compactness inside it. A bounded horizontal lift of a constant target tangent vector $v \in \mathbb{C}$ is

$$V_v = v \frac{\overline{\nabla f_\epsilon}}{\|\nabla f_\epsilon\|^2}, \quad df_\epsilon(V_v) = v.$$

Boundary transversality supplies a smooth horizontal lift tangent to ∂B_r . On a compact collar patch it to V_v by a partition of unity. The patched lift remains horizontal and bounded and is tangent to the sphere. Its flows exist for the required finite times, preserve the sphere and its two sides, and yield compatible local trivializations of the pair (C_t, S_t) . This uses both tameness and transversality, following the bounded-lift construction in [4, pp. 229–230].

Inclusion $j : S_t \hookrightarrow C_t$ preserves intersection numbers. Hence $j_*v = 0$ implies $\langle v, u \rangle = \langle j_*v, j_*u \rangle = 0$ for every u , and local nondegeneracy gives $v = 0$. The global and local homology groups both have dimension μ , so j_* is an isomorphism. The compatible transport above makes it an isomorphism of local systems. The punctured disk contains all critical values and its inclusion in their complement in $\mathbb{C}$ induces an isomorphism on fundamental groups. Thus the global intersection form is nondegenerate and its monodromy is that of this morsification. Rescaling back to q proves the remaining assertions of (G1).

(G2). The Picard–Lefschetz formula is [1, §1.3, p. 26]. At a nondegenerate critical point choose holomorphic coordinates with $q = c_i + uv$, $dx \wedge dy = J du \wedge dv$. The vanishing cycle is $\{uv = t - c_i, |u| = r\}$ and $dx \wedge dy/dq = -J du/u$ on C_t , so the vanishing period is $\mp 2\pi i \sum_\kappa J_{\kappa\kappa}(t - c_i)^\kappa$, holomorphic with value $\mp 2\pi i J(0) \neq 0$ at c_i . Hence $\nu_i \neq 0$, and $T_i \neq 1$ by nondegeneracy of the form.

(G3). By the previous steps $H_1(C_t)$ has the basis $\Delta_1, \dots, \Delta_\mu$, the global monodromy group is generated by the Picard–Lefschetz transformations T_{Δ_i} , and the form is nondegenerate. By the theorem of Gabrielov and Lazzeri [1, Theorem 3.3], the diagram of a singularity relative to any distinguished basis is connected: the graph on $\{1, \dots, \mu\}$ with an edge when $\langle \Delta_i, \Delta_j \rangle \neq 0$ is connected. (The diagram is defined after stabilization, which changes off-diagonal intersection numbers only by signs; self-intersections may change.) Let $W \neq 0$ be invariant. By nondegeneracy there are $w \in W$ and i with $\langle w, \Delta_i \rangle \neq 0$; then $(T_{\Delta_i} - 1)w = \pm \langle w, \Delta_i \rangle \Delta_i$ puts Δ_i in W , and $(T_{\Delta_j} - 1)\Delta_i = \pm \langle \Delta_i, \Delta_j \rangle \Delta_j$ propagates membership along the connected graph. So $W = H_1$. $\square$

For a horizontal family of cycles $\delta(t) \subset C_t$ over a simply connected $U \subset S$ and $\omega = \sum_m z^m \omega_m \in \Omega_{\mathbb{C}[x,y]}^2[z]$ put

$$\pi_{\omega_m}(t) = \int_{\delta(t)} \frac{\omega_m}{dq}, \quad \Pi_\delta(\omega)(t) = \sum_m (-\partial_t)^m \pi_{\omega_m}(t).$$

Lemma 5.2 (Weyl intertwining). $\Pi_\delta(z\omega) = -\partial_t \Pi_\delta(\omega)$, $\Pi_\delta(\partial_z \omega + q\omega) = t \Pi_\delta(\omega)$, and $\Pi_\delta(d_z \beta) = 0$ for $\beta \in \Omega^1[z]$.

Proof. The first is the definition. Since $q = t$ on C_t , $\pi_{q\omega_m} = t\pi_{\omega_m}$, and $(-\partial_t)^m t = t(-\partial_t)^m - m(-\partial_t)^{m-1}$ gives the second. For a polynomial one-form β_0 , $\pi_{dq \wedge \beta_0} = \int_{\delta(t)} \beta_0$ and $\pi_{d\beta_0} = \partial_t \int_{\delta(t)} \beta_0$ (Gelfand–Leray); the two contributions to $\Pi_\delta((d + z dq \wedge) \beta)$ cancel termwise. $\square$

Lemma 5.3 (Period injectivity). Assume (G1), (G3) and $\mu \geq 2$. If $\Pi_\delta(\omega) \equiv 0$ for every δ , then $[\omega] = 0$ in $H_q^2 \otimes \mathbb{C}$.

Proof. If $d\theta = \omega'$ then $d_z \theta = \omega' + z dq \wedge \theta$, so $[\omega'] = -z[dq \wedge \theta]$. Write $\omega = \sum_{m=0}^N z^m \omega_m$ and apply this to the lowest coefficient: subtracting $d_z \theta_0$ removes the z^0 term and changes only the coefficient of z^1 . After N steps $\omega \equiv z^N \omega^\sharp$ modulo $d_z \Omega^1[z]$ with $\omega^\sharp$ independent of z , and $\Pi_\delta(\omega) = (-\partial_t)^N \pi_{\omega^\sharp}$ by Lemma 5.2. Thus every branch of every period of $\alpha^\sharp = \omega^\sharp/dq$ is a polynomial of degree $< N$ (zero when $N = 0$), hence single valued: $\int_{(g-1)\delta} \alpha^\sharp = 0$ for all $g \in \pi_1(S)$. The span of the $(g-1)\delta$ is an invariant subspace, nonzero because an irreducible representation of dimension ≥ 2 is nontrivial; by (G3) it is all of H_1 . So all periods of $\alpha^\sharp$ vanish. Choose $\theta^\sharp$ with $d\theta^\sharp = \omega^\sharp$. Its periods have derivative zero, hence are constant, hence single valued, hence (same argument) zero. So $\theta^\sharp$ is exact along the generic fibres. The critical locus is finite and all fibres are connected and nonempty; these give Bonnet’s primitive and quasi-fibered hypotheses for a map to the line. Thus [3, Definitions 1.3–1.4 and Theorem 1.5] gives $\theta^\sharp = dR + a dq$. Then $\omega^\sharp = da \wedge dq = d_z(a dq)$ and $[\omega] = \pm z^N [\omega^\sharp] = 0$. $\square$

Proposition 5.4 (Irreducibility). Assume (G1)–(G3) and $\mu \geq 2$. Every nonzero element of $H_q^2 \otimes_{F_0} F$ has minimal differential operator of order $\geq \mu$ over F . With Lemma 4.1, $H_q^2 \otimes F$ is an irreducible differential module of rank μ .

Proof. Let $\eta \neq 0$. Multiplying η by an element of $F^\times$ changes neither the differential submodule it generates nor the minimal order of an annihilator (Leibniz), so we may assume $\eta = [\omega]$ with $\omega \in \Omega_{\mathbb{C}}^2[z]$. Let $P = \sum_{l=0}^r p_l(z) \nabla_z^l$, $p_l \in \mathbb{C}[z]$ not all zero, satisfy $P\eta = 0$. Multiplying P on the left by a polynomial in z , which does not change r , we may assume $P(z, \nabla_z)\omega = d_z \beta$ with $\beta \in \Omega^1[z]$. By Lemma 5.2, for every δ ,

$$\hat{P} \Pi_\delta(\omega) = 0, \quad \hat{P} = \sum_l p_l (-\partial_t) t^l = \sum_{n=0}^N a_n(t) \partial_t^n, \quad a_N(t) = (-1)^N \sum_l p_{l,N} t^l,$$

where $N = \max \deg p_l$ and $p_{l,N}$ is the coefficient of z^N in p_l . So $a_N \neq 0$ and $\deg a_N \leq r$. The map $\delta \mapsto \Pi_\delta(\omega)$ from $H_1(C_{t_0}; \mathbb{C})$ to germs at t_0 is linear, compatible with analytic continuation, and nonzero by Lemma 5.3; by (G3) it is injective. If $a_N(c_i) \neq 0$, every solution germ of $\hat{P}$ near c_i is holomorphic at c_i , so continuation around c_i is trivial on the image, contradicting $T_i \neq 1$. Hence a_N vanishes at $c_1, \dots, c_\mu$ and $r \geq \deg a_N \geq \mu$. A proper nonzero submodule would contain an element with minimal operator of order $< \mu$. $\square$

6 The exponential torus and SL_μ

Consider $Y' = AY$ with A as in Lemma 4.1 and let $G \subset \mathrm{GL}_\mu$ be its differential Galois group over F , acting on the solution space V .

Lemma 6.1 (Formal diagonalization). *Let $A = \sum_{j \geq 0} A_j w^j$, $w = 1/z$, with A_0 having distinct eigenvalues $c_1, \dots, c_n$. Then $Y' = AY$ has a formal fundamental matrix $\hat{Y} = \hat{H}(w) z^\Lambda e^{Cz}$ with $C = \mathrm{diag}(c_i)$, Λ a constant diagonal matrix and $\hat{H} \in \mathrm{GL}_n(\mathbb{C}[[w]])$.*

Proof. After a constant change of basis $A_0 = C$. Seek $\hat{H} = \sum_{k \geq 0} H_k w^k$, $H_0 = I$, with $\hat{H}' = A\hat{H} - \hat{H}(C + \Lambda w)$; then $\hat{H} z^\Lambda e^{Cz}$ is a solution. Since $(w^k)' = -kw^{k+1}$, the coefficient of w^k gives, for $k \geq 1$,

$$-(k-1)H_{k-1} = [C, H_k] + \sum_{j=1}^k A_j H_{k-j} - H_{k-1} \Lambda.$$

For $k = 1$ put $\Lambda = \mathrm{diag}(A_1)$; the off-diagonal entries $(c_i - c_l)(H_1)_{il}$ of $[C, H_1]$ determine the off-diagonal part of H_1 . For $k \geq 2$ the diagonal part of the equation reads $-(k-1) \mathrm{diag} H_{k-1} = \mathrm{diag}(A_1^{\mathrm{off}} H_{k-1}^{\mathrm{off}} + \sum_{j=2}^k A_j H_{k-j})$, which determines $\mathrm{diag} H_{k-1}$ from quantities already known, and the off-diagonal part determines H_k^{off} . Induction on k constructs $\hat{H}$. $\square$

Proposition 6.2. *Assume (M), (Q), (G1)–(G3), $\mu \geq 2$. Then $G \supseteq \mathrm{SL}_\mu$.*

Proof. Formal solution and a diagonal subgroup. Let $\mathfrak{K} = \mathbb{C}((w))$ with $' = d/dz = -w^2 d/dw$. By Lemma 4.1(c) and Lemma 6.1, $\hat{H}' = A\hat{H} - \hat{H}(C + \Lambda w)$ with $C = \mathrm{diag}(c_i)$, $\Lambda = \mathrm{diag}(\lambda_i)$, $\hat{H} \in \mathrm{GL}_\mu(\mathbb{C}[[w]])$. Let K_0 be a Picard–Vessiot extension of $\mathfrak{K}$ for the diagonal system $Y' = (C + \Lambda w)Y$; it is generated by nonzero u_i with $u_i' = (c_i + \lambda_i w)u_i$. Then $\hat{Y} = \hat{H} \mathrm{diag}(u_i)$ is a fundamental matrix of $Y' = AY$ in K_0 , and since K_0 has constants $\mathbb{C}$, $\hat{L} = F(\hat{Y}_{ij})$ is a Picard–Vessiot extension of F ; G is its group, written in the basis of columns of $\hat{Y}$. Every $\sigma \in \Gamma = \mathrm{Gal}(K_0/\mathfrak{K})$ multiplies each u_i by a constant, so it restricts to an element of G which is diagonal. Let $D_\Gamma \subset G$ be the resulting closed subgroup of the diagonal torus. A character $x^n = \prod x_i^{n_i}$ is trivial on D_Γ iff $\prod u_i^{n_i}$ is Γ -fixed, i.e. lies in $\mathfrak{K}$. An element $f \in \mathfrak{K}^\times$ has $f'/f \in w\mathbb{C}[[w]]$, while $(\prod u_i^{n_i})'/\prod u_i^{n_i} = \sum n_i c_i + (\sum n_i \lambda_i)w$; so such n satisfy $\sum n_i c_i = 0$. A closed subgroup of a torus is the common kernel of the characters trivial on it, hence

$$D_\Gamma \supseteq T := \{d : d^n = 1 \text{ whenever } \sum n_i c_i = 0\}.$$

T is a torus with character group $\mathbb{Z}^\mu / \{n : \sum n_i c_i = 0\} \cong \sum \mathbb{Z} c_i$, acting on the i th basis vector with weight c_i ; being connected, $T \subset G^0$.

Root decomposition. By (Q) the weights $c_i - c_j$ ($i \neq j$) of T on $\mathfrak{gl}_\mu$ are pairwise distinct and nonzero, so $\mathfrak{g} = \mathrm{Lie} G^0 = (\mathfrak{g} \cap \mathfrak{d}) \oplus \bigoplus_{(i,j) \in R} \mathbb{C} E_{ij}$, $\mathfrak{d}$ the diagonal matrices. G is reductive because V is a faithful irreducible representation (Proposition 5.4), so G^0 is reductive; $\mathfrak{g} \cap \mathfrak{d}$ is the centralizer of $\mathrm{Lie} T$, a Cartan subalgebra, and therefore $R = -R$. Also R is closed under $(i, j), (j, l) \mapsto (i, l)$. So R defines an equivalence relation with classes $I_1, \dots, I_s$; each $V_{I_a} = \mathrm{span}\{e_i : i \in I_a\}$ is an irreducible G^0 -module and they are pairwise nonisomorphic, so the G^0 -submodules of V are exactly the sums of the V_{I_a} .

One class. Since G^0 is normal, G permutes the V_{I_a} . The fixed field E_0 of G^0 in $\hat{L}$ is a finite Galois extension of F with group G/G^0 . Under the isomorphism of $\hat{L}$ with the Picard–Vessiot field generated by an analytic fundamental matrix, its elements are algebraic functions meromorphic on the universal cover of $\mathbb{C}^\times$ (the system is singular only at $0, \infty$), so $E_0 = F(\zeta)$ with $\zeta^d = z$. The image of D_Γ in $\mathrm{Gal}(E_0/F)$ has fixed field $E_0 \cap \mathfrak{K}$. This is $F(\zeta^{d'/d})$ for some $d' \mid d$, and

$(\zeta^{d/d'})^{d'} = w^{-1}$ forces $d' = 1$ because w -orders in $\mathfrak{K}$ are integers. So D_Γ maps onto G/G^0 , i.e. $G = G^0 D_\Gamma$. Diagonal matrices preserve each V_{I_a} , so each V_{I_a} is G -stable and irreducibility gives $s = 1$. Then every E_{ij} lies in $\mathfrak{g}$ and $\mathrm{SL}_\mu \subset G^0$. $\square$

7 Face disjointness

Stable lattices at infinity. Put $R_\infty = \mathbb{C}[[w]]$, $K_\infty = \mathbb{C}((w))$ and $\delta = d/dz = -w^2 d/dw$. A lattice in a differential module over K_∞ is a finite free R_∞ -submodule spanning it. A lattice L is stable if $\partial L \subset L$. Since $\delta(R_\infty) \subset wR_\infty$, its operator induces a $\mathbb{C}$ -linear operator on L/wL , called the leading operator of this stable lattice. In the coefficient-module convention its matrix for a system $Y' = E(w)Y$ is $E(0)^\top$, which has the same spectrum as $E(0)$.

Lemma 7.1 (Stable lattices and subquotients). *Let $L \subset M$ be a stable lattice and $N \subset M$ a differential submodule. Then $L_N = L \cap N$ and $L_{M/N} = \mathrm{im}(L \rightarrow M/N)$ are stable lattices, and*

$$0 \longrightarrow L_N/wL_N \longrightarrow L/wL \longrightarrow L_{M/N}/wL_{M/N} \longrightarrow 0$$

is exact and compatible with the leading operators. Their characteristic polynomials therefore multiply. In particular a direct sum of successive quotients of a differential-module filtration has a stable lattice with the same leading characteristic polynomial as L .

Proof. The intersection is a saturated, finitely generated submodule of L : if $wv \in L \cap N$ and $v \in L$, then $v \in N$ because N is a K_∞ -vector space. The intersection and the quotient are free over the discrete valuation ring R_∞ and span the desired spaces. Stability is inherited. Since the quotient is free, reduction modulo w preserves exactness. Also $\delta(w) = -w^2$, so each leading operator is well-defined and the sequence intertwines them. Apply this successively to the filtration. $\square$

Lemma 7.2 (Detection of a rank-one leading coefficient). *If a module M with stable lattice L contains a nonzero vector v with $\partial v = (d + \rho w)v$, $d, \rho \in \mathbb{C}$, then d is an eigenvalue of the leading operator on L/wL .*

Proof. The saturated lattice $L \cap K_\infty v$ has a generator fv , where $f = w^m u$ with $u \in R_\infty^\times$. Its coefficient is

$$d + \rho w + \frac{\delta f}{f}, \quad \frac{\delta f}{f} = -mw - w^2 \frac{u_w}{u} \in wR_\infty.$$

Its leading eigenvalue is therefore d . Its reduction injects into L/wL by Lemma 7.1. $\square$

Corollary 7.3 (A leading-spectrum obstruction). *Suppose M has a stable lattice with leading spectrum Σ . Let P admit a basis over K_∞ with $\partial e_i = (c_i + \lambda_i w)e_i$. If*

$$\mathrm{End}^0(P) \hookrightarrow \mathrm{End}(M^{\mathrm{ss}})$$

is a differential-module embedding, then $c_i - c_j \in \Sigma - \Sigma$ for all $i \neq j$.

Proof. By Lemma 7.1, M^{ss} has a stable lattice L^{ss} with the same leading characteristic polynomial. The lattice $\mathrm{End}_{R_\infty}(L^{\mathrm{ss}})$ is stable; its leading operator is the commutator with that of L^{ss} , with eigenvalues $s - s'$ for $s, s' \in \Sigma$. The vector $e_i \otimes e_j^*$ is trace-free for $i \neq j$ and has differential coefficient $c_i - c_j + (\lambda_i - \lambda_j)w$. Its image in the endomorphism module is nonzero, so Lemma 7.2 gives the result.

If M^{ss} was taken over F first, extend a composition series from F to K_∞ and apply the lattice lemma to this filtration. Its quotients need not remain simple. No assertion that semisimplification commutes with this extension is used. $\square$

For a face polynomial p of degree $d \geq 2$, the one-variable reduction gives a matrix in $M_{d-1}(k[z^{-1}])$ whose leading matrix is multiplication by p on $k[s]/(p')$. Thus its stable lattice has as leading spectrum the critical values of p , also when those values repeat. For the interior module, Lemma 6.1 supplies the diagonal basis required in Corollary 7.3. These are all the local facts needed below; no general formal-normal-form theorem is required.

Let $\mathcal{B}$ be the closed homogeneous system satisfied by the boundary vector (reduced face moments, endpoint exponentials, 1), $\mathcal{T}_\partial$ its tensor category and G_∂ its Galois group.

Proposition 7.4. *Suppose $G \supseteq \mathrm{SL}_\mu$, $\mu \geq 2$, and that the interior module has the formal diagonalization of Lemma 6.1. If $\mathrm{End}^0(A)$ belongs to $\mathcal{T}_\partial$, then there is a face e with $d_e \geq 2$ such that every $c_i - c_j$ is a difference of two critical values of p_e . In particular (F) implies $\mathrm{End}^0(A) \notin \mathcal{T}_\partial$.*

Proof. The Galois group of $\mathrm{End}^0(A)$ is PGL_μ , so there is a surjection $\rho : G_\partial \rightarrow \mathrm{PGL}_\mu$ through which G_∂ acts on $\mathfrak{sl}_\mu$ by $\mathrm{Ad} \circ \rho$. The unipotent radical dies, so ρ factors through $G' = \mathrm{Gal}(\mathcal{B}^{\mathrm{ss}}) \subset \prod_e \mathrm{GL}(W_e) \times (\text{torus})$, W_e the solution space of $H_{p_e}^{\mathrm{ss}}$. Write $\mathrm{Lie} G' = \mathfrak{z} \oplus \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_r$. Since $\mathfrak{sl}_\mu$ is simple, $\ker d\rho$ contains $\mathfrak{z}$ and all simple ideals but one, $\mathfrak{s}$; as $\ker d\rho$ is $\mathrm{Ad}(G')$ -stable, so is $\mathfrak{s}$, and $d\rho : \mathfrak{s} \rightarrow \mathfrak{sl}_\mu$ is a G' -equivariant isomorphism. The projections $\mathfrak{s} \rightarrow \mathfrak{gl}(W_e)$ are G' -equivariant and each is injective or zero; they are not all zero because the remaining factor is abelian. For an e with injective projection, the G_∂ -module $\mathfrak{sl}_\mu$ embeds in $\mathrm{End}(W_e)$. Using the covariant horizontal-fibre equivalence and the trace dualities noted above, this gives a differential-module embedding

$$\mathrm{End}^0(A) \hookrightarrow \mathrm{End}(H_{p_e}^{\mathrm{ss}}).$$

For the face module the leading spectrum is the set of critical values of p_e . Corollary 7.3, after extension to K_∞ , therefore makes every $c_i - c_j$ a difference of two such critical values. This contradicts (F). $\square$

8 The saddle lemma

Let $\mathcal{N} = \mathrm{span}_F\{1, \text{reduced face moments, endpoint exponentials}\}$.

Lemma 8.1. *Assume (G1)–(G3), $\mu \geq 2$, and (S) or (S*). Then $I_D(dx \wedge dy) \notin \mathcal{N}$.*

Proof. Suppose $P(z)I_D(dx \wedge dy) = \sum_j R_j(z)g_j$ with $0 \neq P$, $R_j \in \mathbb{C}[z]$ and g_j among the generators. The inverse Laplace transforms of the g_j have algebraic densities on intervals free of the finitely many face-critical, vertex and zero values, and point masses elsewhere. The compactly supported distributions are determined by their Laplace transforms: restrict an identically zero entire transform to the imaginary axis and apply Fourier uniqueness. Hence on such an interval the density ρ of $I_D(dx \wedge dy)$ satisfies: $g = P(-\partial_t)\rho$ is algebraic over $\mathbb{C}(t)$. Under (S) this applies to $(c, c + \varepsilon)$ for small $\varepsilon > 0$.

Local form under (S). By the real-analytic Morse lemma there are local coordinates with $q = c + uv$, $dx \wedge dy = J du \wedge dv$, $J(0)^2 = |\det \mathrm{Hess} q(P_0)|^{-1}$. Take a box $\{|u|, |v| \leq \delta\}$ on which the multiplicity of D is w . After shrinking δ the edges of the box are transverse to the levels near c and its corners have levels $c \pm \delta^2 \neq c$. Outside the open box the level curves near level c meet neither a critical point (distinct critical values) nor a tangency or vertex of ∂D (by (S)), so that contribution to ρ is analytic at c . Inside, with $s = t - c$,

$$\rho_{\mathrm{box}}(s) = \pm w \int_{|s|/\delta \leq |u| \leq \delta} J(u, s/u) \frac{du}{|u|} = \pm w(-2j(s) \log s + \text{analytic}) \quad (0 < s < \delta^2),$$

where $J = \sum J_{\kappa\lambda} u^\kappa v^\lambda$ and $j(s) = \sum_\kappa J_{\kappa\kappa} s^\kappa = \frac{1}{2\pi i} \oint_{|u|=r} J(u, s/u) \frac{du}{u}$; the terms with $\kappa \neq \lambda$ integrate to convergent power series in s . Since $dx \wedge dy/dq = -J du/u$ on C_t , j is $\mp \frac{1}{2\pi i}$ times the period $h(t) = \int_{\nu(t)} dx \wedge dy/dq$ over the vanishing cycle, and $j(0) = J(0) \neq 0$. Thus ρ continues around c with variation a nonzero constant times h , which is (S^*) .

Conclusion from (S^) .* Write the variation as λh , $\lambda \neq 0$. The vanishing period h is fixed by the local monodromy, so N turns around c_i change ρ by $N\lambda h$. The germ g continues with ρ and remains algebraic, so for some $N \geq 1$ it returns to itself; therefore $N\lambda P(-\partial_t)h = 0$ near c_i and h extends to an entire function. So h is single valued on S , i.e. $\int_{(g-1)\nu_i} dx \wedge dy/dq = 0$ for all $g \in \pi_1(S)$. As in Lemma 5.3 the $(g-1)\nu_i$ span H_1 ($\nu_i \neq 0$ is not invariant, by (G3) and $\mu \geq 2$), so the period over ν_i itself vanishes, contradicting (G2). $\square$

9 Proof of Theorem 1.2

(i) is Lemma 4.1. For (ii): integrating the certificates of qm gives $M' = AM + b$ with $b_m = \int_{\partial D} e^{zq} (P_{qm} dy - Q_{qm} dx) \in \mathcal{N}$. The actual one-column criterion, Theorem 3.3, requires: (1) $G \supseteq \text{SL}_\mu$, which is Proposition 6.2; (2) $\text{End}^0(A) \notin \mathcal{T}_\partial$, which is Proposition 7.4 with (F) and Lemma 6.1; (3) some $M_m \notin \mathcal{N}$, which is Lemma 8.1 since $1 \in B$. Hence the M_m are algebraically independent over $F(\mathcal{N}) = K_\partial$.

(iii) The vector (boundary coordinates, M) consists of E -functions (Theorem 2.1) and satisfies a homogeneous linear system over $k(z)$ whose only finite pole is $z = 0$: for M by Lemma 4.1, for the faces by its one-variable case. Let d be the transcendence degree of $K_{\partial,0}$ over $k(z)$. Part (ii) gives transcendence degree $d + \mu$ for the combined vector. Beukers' form of Siegel–Shidlovskii [2, Theorem 1.1, p. 370], applied first to the boundary vector and then to the combined vector, gives the numerical degrees d and $d + \mu$ at every $\xi \in k^\times$. Their difference is μ . The one-variable normal forms have only powers of z as denominators, so these reduced boundary values generate the stated face-value field at all such ξ .

(iv) Specialize (i) at $z = \xi$: the boundary term lies in $K_{\partial,\xi}$, so algebraicity forces all $a_{r,m}(\xi) = 0$ by (iii), which gives the exactness; conversely exactness gives membership by Stokes. $\square$

10 The rank-six example

Let $q = x^4 + y^3 + xy - y$, so $(a, b) = (4, 3)$, $\mu = 6$, $B = \{1, x, x^2, y, xy, x^2y\}$, and D the triangle of Example 1.5. The following are exact computations (script supplied).

- *Connection.* $A = A_0 + A_1/z$ with $\det(t - A_0)$ proportional to

$$\chi(t) = 1289945088 t^6 - 573308928 t^4 + 83980800 t^3 + 83854656 t^2 + 5477429 t - 4197429.$$

The same polynomial is obtained independently as the resultant eliminating the critical points.

- *(M) and (Q).* χ is irreducible over $\mathbb{Q}$; modulo 7 it factors with degrees (1, 5) and modulo 31 with degrees (1, 2, 3). A transitive group of degree 6 containing a 5-cycle is primitive, and with a transposition (the cube of the second Frobenius) it is S_6 . Given a relation $\sum a_i c_i = 0$ with rational a_i , applying a transposition and subtracting yields $(a_i - a_j)(c_i - c_j) = 0$. Thus all a_i are equal; a relation among the differences has coefficient sum zero and is trivial.
- *(S).* The critical points are $x = 1 - 3y^2$ with $-108y^6 + 108y^4 - 36y^2 + y + 4 = 0$. Write $f(y)$ for this sextic. Sturm counts give exactly two real roots, one in $(-\frac{420}{1000}, -\frac{419}{1000})$ and one in

$(\frac{722}{1000}, \frac{723}{1000})$. On the critical locus $\det \text{Hess } q = 72x^2y - 1 = -f'(y)$, and f' has no root in either interval, with $f' > 0$ on the first and $f' < 0$ on the second. So the first root gives a saddle $P_0 \approx (0.471644, -0.419665)$, $c = q(P_0) \approx 0.197305$, and the second does not. The polynomials $x - \frac{y+1}{2}$ and $1 - \frac{y+1}{2} - x$ with $x = 1 - 3y^2$ have no root on the first interval and are positive there, so P_0 is interior to D .

- (F) and the rest of (S) hold by Corollary 1.4 with $k_0 = \mathbb{Q}$, since $[\mathbb{Q}(c) : \mathbb{Q}] = 6 \geq 4$.

For reproducibility, writing $\bar{\chi}_p$ for the monic reduction modulo p ,

$$\begin{aligned}\bar{\chi}_5 &= t^6 - t^4 + 2t^2 - 2t + 2, \\ \bar{\chi}_7 &= (t+3)(t^5 - 3t^4 + 3t^2 + 3t + 2), \\ \bar{\chi}_{31} &= (t-14)(t^2 + 8t + 10)(t^3 + 6t^2 - 14t + 1).\end{aligned}$$

The displayed factors modulo 7, 31 are irreducible and squarefree. Irreducibility modulo 5 is certified by

$$\gcd(\bar{\chi}_5, t^{5^2} - t) = \gcd(\bar{\chi}_5, t^{5^3} - t) = 1, \quad t^{5^6} \equiv t \pmod{\bar{\chi}_5}.$$

All these are exact polynomial computations in the accompanying certificate. The supplementary exact checks give $\text{tr}(A_0^k(A_1 + I)) = 0$ for $0 \leq k \leq 5$. Since A_0 has simple spectrum, the Vandermonde matrix in its eigenvalues shows that all six diagonal entries of A_1 in its eigenbasis are -1 . Thus the six powers in Lemma 6.1 are exactly -1 .

The exact checks are run by `make test`. No numerical quadrature or asymptotic fitting is used to certify the hypotheses.

11 The elliptic triangle and the continuation alternative

Fix the counterclockwise triangle

$$D = \text{conv}\{(-1/2, -1), (1/2, 0), (-1/2, 1)\} = \{-1/2 \leq x \leq 1/2, |y| \leq 1/2 - x\},$$

and

$$q = y^2 - x^3 + 3x, \quad M_j = \int_D x^j e^{zq} dx dy \quad (j = 0, 1).$$

A polynomial parametrization is $\Phi(u, v) = (u - 1/2, (1 - u)(2v - 1))$, with Jacobian $2(1 - u)$.

For $\beta = P dy - Q dx$, one has

$$d_z \beta = (L_x P + L_y Q) dx \wedge dy, \quad L_x = \partial_x + 3z(1 - x^2), \quad L_y = \partial_y + 2zy.$$

The operators act in separate variables. Their one-variable quotients have bases $1, x$ and 1 , respectively; both operators are injective on polynomials. Tensoring the two one-variable complexes proves

$$H_q^2 = F_0[dx \wedge dy] \oplus F_0[x dx \wedge dy]. \quad (11.1)$$

In particular the rank is exactly two, not merely at most two.

The explicit reduction recurrences are

$$[x^i y^j] = -\frac{j-1}{2z} [x^i y^{j-2}] \quad (j \geq 1), \quad (11.2)$$

$$[x^{m+2}] = [x^m] + \frac{m}{3z} [x^{m-1}] \quad (m \geq 0), \quad (11.3)$$

with the zero-coefficient terms omitted. Each use records a polynomial primitive. Hence every $r \in k[x, y]$ has a unique remainder $a_r(z) + b_r(z)x$, with $a_r, b_r \in k[z, z^{-1}]$, and an identity

$$r = a_r + b_r x + L_x P_r + L_y Q_r, \quad P_r, Q_r \in k[z, z^{-1}, x, y]. \quad (11.4)$$

Let

$$C = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -5/6 & 0 \\ 0 & -7/6 \end{pmatrix}, \quad A = C + D_0/z.$$

The following are identities of polynomial forms, with Laurent coefficients in z :

$$q \, dx \wedge dy = -\frac{5}{6z} \, dx \wedge dy + 2x \, dx \wedge dy + d_z \beta_0, \quad (11.5)$$

$$q \, dx \wedge dy = 2 \, dx \wedge dy - \frac{7}{6z} \, dx \wedge dy + d_z \beta_1, \quad (11.6)$$

where

$$\beta_0 = \frac{x}{3z} \, dy - \frac{y}{2z} \, dx, \quad \beta_1 = \frac{x^2 - 2}{3z} \, dy - \frac{xy}{2z} \, dx.$$

Define actual face integrals

$$B_{x,j} = \int_{\partial D} x^j e^{zq} \, dy, \quad B_{y,j} = - \int_{\partial D} x^j y e^{zq} \, dx.$$

Then

$$\begin{pmatrix} M_0 \\ M_1 \end{pmatrix}' = \begin{pmatrix} -5/(6z) & 2 \\ 2 & -7/(6z) \end{pmatrix} \begin{pmatrix} M_0 \\ M_1 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} B_{y,0}/2 + B_{x,1}/3 \\ B_{y,1}/2 + B_{x,2}/3 - 2B_{x,0}/3 \end{pmatrix}. \quad (11.7)$$

This is obtained by integrating (11.6), not by choosing an isomorphic abstract connection and then normalizing a solution.

11.1 A finite actual boundary field

Put

$$p(x) = -x^3 + x^2 + 2x + \frac{1}{4}, \quad P_j(z) = \int_{-1/2}^{1/2} x^j e^{zp(x)} \, dx, \quad Q_0(z) = \int_{-1}^1 e^{z(y^2 - 11/8)} \, dy,$$

and $E_a = e^{-3z/8}$, $E_b = e^{11z/8}$. Parametrizing the three oriented edges gives

$$B_{x,j} = 2P_j - (-1/2)^j Q_0, \quad B_{y,j} = P_j - 2P_{j+1}. \quad (11.8)$$

The one-variable reductions imply

$$K_{\partial,0} = k(z)(P_0, P_1, Q_0, E_a, E_b). \quad (11.9)$$

Indeed the two slanted faces have the same phase p ; their arbitrary polynomial amplitudes reduce to P_0, P_1 and endpoints. The vertical quadratic amplitudes reduce to Q_0 and endpoints.

For completeness, a closed boundary system is

$$P'_0 = \left(\frac{17}{36} - \frac{1}{3z} \right) P_0 + \frac{14}{9} P_1 + \frac{E_b + 5E_a}{18z}, \quad (11.10)$$

$$P'_1 = \left(\frac{28}{27} + \frac{1}{9z} \right) P_0 + \left(\frac{163}{108} - \frac{2}{3z} \right) P_1 + \frac{-53E_b + 41E_a}{108z}, \quad (11.11)$$

$$Q'_0 = \left(-\frac{11}{8} - \frac{1}{2z} \right) Q_0 + \frac{E_a}{z}, \quad E'_a = -\frac{3}{8} E_a, \quad E'_b = \frac{11}{8} E_b. \quad (11.12)$$

For example, $P_2 = (2P_1 + 2P_0)/3 - (E_b - E_a)/(3z)$, so (11.7) closes with this boundary vector. The only finite pole of the combined rational system is zero.

11.2 The actual normalization at zero

The face forcing in (11.7) is $b(z)/z$ with b entire. Writing $M = \sum v_m z^m$, $b = \sum b_m z^m$, its holomorphic solution obeys

$$(mI - D_0)v_m = Cv_{m-1} + b_m, \quad v_{-1} = 0.$$

Every $mI - D_0$, $m \geq 0$, is invertible. Thus there is at most one holomorphic solution at zero with these actual forcing functions. The integral supplies it, with

$$M_0(0) = 1, \quad M_1(0) = -\frac{1}{6}.$$

This supplies a second, independent check that no unspecified homogeneous solution has been added to the actual column.

11.3 The actual continuation

On the regular phase fibre $y^2 = x^3 - 3x + t$, the convention $dq \wedge \alpha = dx \wedge dy$ gives $\alpha = -dx/(2y)$.

Lemma 11.1 (Continuation for the elliptic triangle). *The density germ of M_0 just to the right of $t = 0$ satisfies (S^*) .*

Proof. We verify a nonzero elliptic monodromy variation of the actual density germ. Put $f(x) = -x^3 + 3x$, let $r_2(s) \in (-1, 1)$ be its middle inverse branch for $-2 < s < 2$, and let $c(t)$ be the inverse branch of p beginning at $c(0) \in (-1/5, 0)$. Set $a(t) = 1/2 - c(t)$. Coarea on the triangle gives, near zero,

$$\rho(t) = \int_{-a(t)}^{a(t)} \frac{dy}{3(1 - r_2(t - y^2)^2)}. \quad (11.13)$$

Here $p'(x) = -3x^2 + 2x + 2$ is positive on $[-1/5, 1]$. Moreover $p(-1/5) < 0 < p(0)$ and $p(4/5) < 2 < p(1)$, so $c(t)$ is a single real-analytic branch for $0 \leq t \leq 2$, with $c(0) \in (-1/5, 0)$ and $c(2) \in (4/5, 1)$. For $|y| \leq |a(t)|$ one has

$$f(c(t)) = t - a(t)^2 \leq t - y^2 \leq t < 2, \quad f(c(t)) > -2.$$

Thus the middle inverse branch is regular on this moving segment for $0 < t < 2$. Writing the signed integral on the fixed interval as

$$a(t) \int_{-1}^1 \frac{ds}{3(1 - r_2(t - a(t)^2 s^2)^2)}$$

proves real analyticity and hence local holomorphic continuation along that interval. At $t = 11/8$ one has $c(t) = 1/2$, so the endpoints meet, but the displayed expression is analytic and zero there. This specifies the continuation through that otherwise excluded boundary value. Near $t = 2$, $a(t) < 0$ and $a(2) \neq 0$, and both endpoints are unramified.

Write $\varepsilon = 2 - t$ and $r_2(t - y^2) = 1 - v$. Since $f(1 - v) = 2 - 3v^2 + v^3$, near the node the integrand is

$$\frac{1}{2\sqrt{3}\sqrt{\varepsilon + y^2}} + O(1).$$

Choose a fixed $b > 0$ smaller than $|a(2)|$. The outer parts $|y| \geq b$ extend holomorphically to $t = 2$; on $|y| \leq b$ the remainder in the displayed expansion is uniformly bounded for $\varepsilon > 0$. Since the integration orientation is reversed near 2, it follows that

$$\rho(t) = \frac{1}{2\sqrt{3}} \log(2 - t) + O(1) \quad (t \rightarrow 2^-). \quad (11.14)$$

The local geometry makes the corresponding variation precise. Put $v = 1 - x$ and $u = v\sqrt{3 - v}$. Then $q - 2 = y^2 - u^2$ near the node, so a nearby local fiber is an annulus. The path outside a small nodal neighborhood and its two endpoints return under a sufficiently small loop. Its variation lies in the one-dimensional cycle space of that annulus. The coefficient in (11.14) proves it is nonzero. The local core cycle δ_+ is fixed by this nodal monodromy (equivalently by its Dehn twist). Consequently

$$(T_+ - I)\rho = c_* h_+(t), \quad c_* \neq 0, \quad h_+(t) = \int_{\delta_+} \frac{dx}{2y}, \quad T_+ h_+ = h_+.$$

This is also the local annulus proof of the Picard–Lefschetz formula in this case; the nonzero coefficient has just been checked on the actual path by (11.13).

The variation is a nonzero constant multiple of the required Gelfand–Leray vanishing period; the common sign is immaterial. $\square$

Remark 11.2. This is continuation of a regular density germ, not a claim that the original compactly supported density is nonzero outside $q(D) = [-11/8, 11/8]$.

11.4 Verification of the comparison hypotheses

The phase has $(a, b) = (3, 2)$ and its two critical values are $-2, 2$. Thus (M) holds and (Q) is automatic. For a slanted edge put $\beta = 7\sqrt{7}/27$. The substitution $x = 1/3 + (\sqrt{7}/3)s$ gives

$$p(x) = \frac{107}{108} + \beta(-s^3 + 3s).$$

Its critical values are $107/108 \pm 2\beta$, so their nonzero differences are $\pm 4\beta$, whereas the interior differences are ± 4 . Since $\beta^2 = 343/729 \neq 1$, these do not agree. The vertical quadratic face has only one critical value, so it also satisfies (F). Lemma 11.1 supplies (S^*) , although the saddle $(1, 0)$ lies outside the triangle and (S) fails.

Corollary 11.3 (Elliptic functional and numerical comparison). *For this triangle and phase, M_0, M_1 are algebraically independent over $\mathbb{C}(z)(P_0, P_1, Q_0, E_a, E_b)$. At every nonzero algebraic ξ , their values are algebraically independent over the corresponding fixed-face value field. Every polynomial amplitude has the unique remainder (11.4), and its value is algebraic over the face-value field exactly when its specialized polynomial twisted class is zero.*

Proof. Apply Theorem 1.2, using the verifications above. The formulas (11.6)–(11.9) retain the actual normalization and the finite generating boundary vector. Thus this is also a specialization of the general proof, not just an abstract identification of two homogeneous equations. $\square$

12 Scope, dependencies, and verification

The normal-form theorem is algebraic and the polynomial boundary-system construction allows arbitrary polynomial phases. Algebraic independence requires the explicitly stated hypotheses of Theorem 1.2, verified for the two examples above. The result is not contingent on an unproved genericity conjecture: (Q) is a hypothesis on the specified critical values, certified in the examples. Its failure alone does not imply that the homogeneous Galois group is smaller.

All boundary fields are attached to the finitely many fixed faces and endpoints of the chain. Replacing them by the field of all compact polynomial-line periods is not an asserted extension.

Only one phase and one actual column are considered. No several-chain or multiblock formal completeness is inferred. Numerical independence is a statement after specializing the parameter; it is not injectivity of evaluation on the unspecialized algebra containing $z - \xi$.

External inputs. The geometric proof uses Broughton [4, Definition 3.1, Theorem 1.2, pp. 229–231] and the Picard–Lefschetz, distinguished-basis, connectedness, and nondegeneracy statements in [1, §1.3; §2.1; Theorem 3.3; pp. 100, 411]. The needed perturbation and compatible-transport arguments are included in Proposition 5.1. Polynomial relative exactness uses [3, Definitions 1.3–1.4, Theorem 1.5]. Ordinary Picard–Vessiot theory and its tensor equivalence use [10, Theorem 1.5.2], and the complete reducibility in the actual criterion uses [7, Proposition 15.15, Corollary 22.43]. The general finiteness assertion uses the explicitly identified Proposition 9.1 of Sabbah [9]. Numerical specialization uses [2, Theorem 1.1] separately for the boundary and combined vectors. Neither Wasow’s formal simplification nor a general formal-normal-form theorem is needed: the relevant simple-spectrum and lattice statements are proved in the text.

Computational verification. The repository contains the retained exact elliptic-amplitude certificate, the rank-six polynomial/resultant/Frobenius/Sturm certificate, and exact checks of the formal recursion and six formal powers. The rank-six geometric inequalities are certified on whole rational intervals. These tests verify finite calculations, not the general monodromy or Galois proofs. Compilation, regression tests, and AI-assisted audits are not independent specialist refereeing or formal verification.

AI assistance and responsibility. OpenAI ChatGPT and Anthropic Claude were used for proof exploration, drafting, symbolic checks, and adversarial review. They are not authors. The authors are responsible for the statements, proofs, citations, and remaining errors.

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